MediXAI: Explainable Unified Multi-Disease Prediction System Using NHANES Data

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Submitted By

NAME : SAMBODH GUPTA

SEMESTER: VI

ROLL NO: 230101019

SUPERVISED BY

DR. KISHORJIT NONGMEIKAPAM

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY SENAPATI, MANIPUR

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Declaration

This project titled **"MediXAI: Explainable Unified Multi-Disease Prediction System Using NHANES Data"** represents my original work and reflects my deep interest in the fields of machine learning, healthcare, and transparent AI systems. The concepts, design, implementation, and documentation presented herein have been developed and executed solely by me. Any ideas, references, datasets, or materials derived from external sources — including the NHANES dataset, the UCI Heart Disease dataset, the Oxford Parkinson's Voice dataset, and related academic literature — have been duly acknowledged and cited in accordance with established academic standards.

I affirm that this submission upholds the principles of academic honesty and integrity. No part of this work has been plagiarised, misrepresented, or falsified.

DATE: 22-04-2026

Sambodh Gupta
230101019

Department of Computer Science and Engineering
Indian Institute of Information Technology Senapati, Manipur

Dr. Kishorjit Nongmeikapam
Associate Professor, CSE

Email: kishorjit@iiitmanipur.ac.in
Contact No: +91 8974008610

To Whom It May Concern

This is to certify that the project report entitled **"MediXAI: Explainable Unified Multi-Disease Prediction System Using NHANES Data,"** submitted to the Department of Computer Science and Engineering, IIIT Senapati, Manipur, in partial fulfilment for the award of the degree of Bachelor of Technology in Computer Science and Engineering, is a record of bonafide work carried out by **Sambodh Gupta**, bearing roll number 230101019.

Signature of Supervisor

Dr. Kishorjit Nongmeikapam

Signature of the Examiner 1 .....

Signature of the Examiner 2 .....

Signature of the Examiner 3 .....

Signature of the Examiner 4 .....

Abstract

This project presents **MediXAI: Explainable Unified Multi-Disease Prediction System** that integrates three independent clinical machine learning models into a single, accessible, and interpretable web platform. The system addresses a critical gap in modern healthcare informatics: the lack of transparency and clinical trust in black-box predictive models.

MediXAI covers three diseases of significant global burden — Diabetes, Heart Disease, and Parkinson's Disease — each modelled using a carefully selected machine learning algorithm trained on nationally and internationally recognised clinical datasets. The Diabetes module employs a Random Forest classifier trained on the NHANES dataset using seven clinical features, achieving a test accuracy of 95.85% and a ROC-AUC of 0.9761. The Heart Disease module uses an ExtraTrees classifier trained on the UCI Heart Disease dataset (89.13% accuracy, ROC-AUC 0.97), selected as the best-performing model from seventeen evaluated algorithms. The Parkinson's Disease module employs an XGBoost classifier trained on the Oxford Voice dataset, achieving approximately 95% accuracy and a ROC-AUC of 0.98.

All three models are augmented with Explainable AI (XAI) techniques. SHAP (Shapley Additive Explanations) provides mathematically rigorous, feature-level attribution grounded in cooperative game theory, while LIME (Local Interpretable Model-agnostic Explanations) offers locally faithful surrogate explanations for individual predictions. Together, these frameworks transform opaque model outputs into human-readable visualisations that clinicians and patients can understand and act upon.

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Chapter 1

Introduction

Artificial Intelligence (AI) and Machine Learning (ML) are reshaping the modern landscape of healthcare by enabling systems that can learn from vast amounts of medical data and assist in making evidence-based clinical decisions. From early disease detection to patient risk stratification, these technologies have proven capable of complementing human expertise with data-driven insights.

Despite substantial progress in predictive modelling, the integration of AI in healthcare continues to face three persistent challenges: lack of transparency, limited interpretability, and poor generalisability of black-box algorithms. In medical contexts, interpretability is not merely a desirable feature — it is an ethical requirement.

1.1 Overview of MediXAI

MediXAI (Medical Explainable AI) is a unified multi-disease prediction platform that brings together three independent clinical machine learning models under a single, accessible, and interpretable web interface. The system addresses a fundamental gap in the current landscape of medical AI: most existing tools are either single-disease systems, lack explainability, or require technical expertise to operate.

MediXAI is designed to serve both healthcare professionals and the general public, offering real-time risk predictions for Diabetes, Heart Disease, and Parkinson's Disease. The platform rests on three foundational pillars: **predictive accuracy**, **explainability**, and **accessibility**. Figure 1.1 shows the home dashboard of the deployed application.

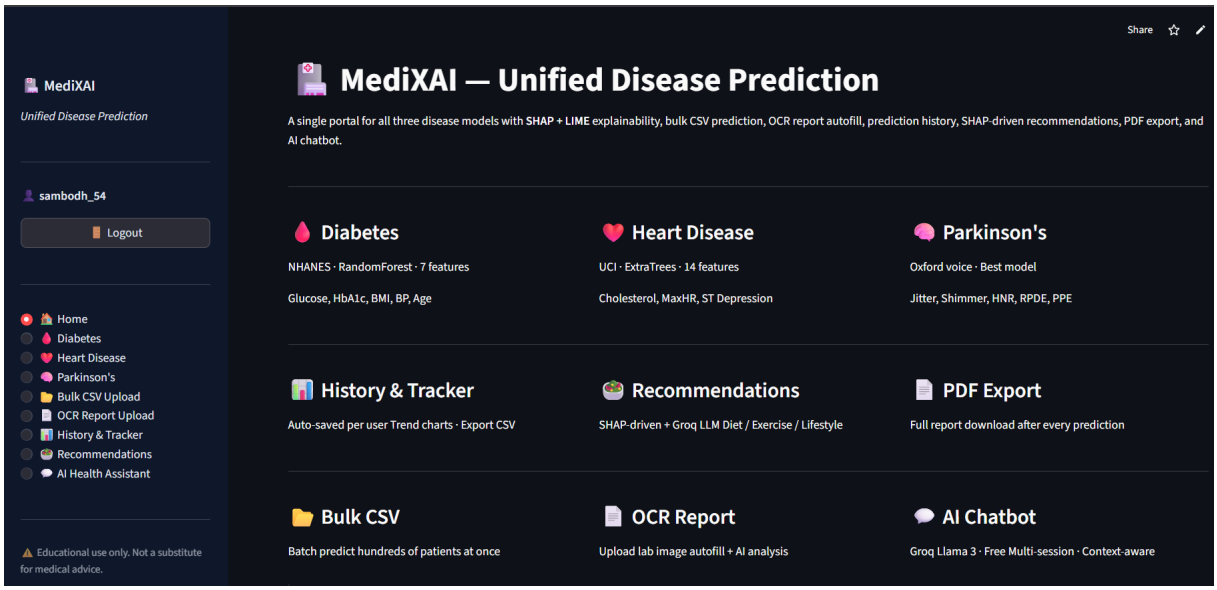


Figure 1.1: MediXAI home dashboard showing the three disease prediction modules (Diabetes, Heart Disease, Parkinson's), along with integrated features including Bulk CSV upload, OCR Report analysis, History Tracker, Recommendations, PDF Export, and the AI Health Assistant chatbot.

1.2 Rationale and Design Philosophy

The design of MediXAI is grounded in a set of guiding principles derived from the intersection of software engineering best practices, clinical informatics standards, and responsible AI development.

Human-Centred Design: Every interface component is designed with the end-user in mind. All technical outputs — probabilities, SHAP values, LIME weights — are translated into plain-language risk assessments and visual summaries.

Transparency by Default: Unlike systems that treat explainability as an optional add-on, Medi XAI enforces XAI at the architectural level. Every prediction is accompanied by both SHAP and LIME explanations.

Modularity and Extensibility: The system is architected so that each disease module, utility function, and interface page is independently maintainable. Adding a new disease module requires no modification to existing code.

Ethical Responsibility: The system is explicitly designed not to replace clinical judgement. All predictions carry a disclaimer affirming that outputs are intended for educational and screening purposes only.

1.3 Development Methodology

The development process followed a structured, iterative model inspired by the Software Development Life Cycle (SDLC), progressing through six phases: literature review, dataset curation and preprocessing, model development and benchmarking, explainability integration, system integration and interface design, and testing and optimisation.

Figure 1.2 illustrates the project timeline across a three-month development period.

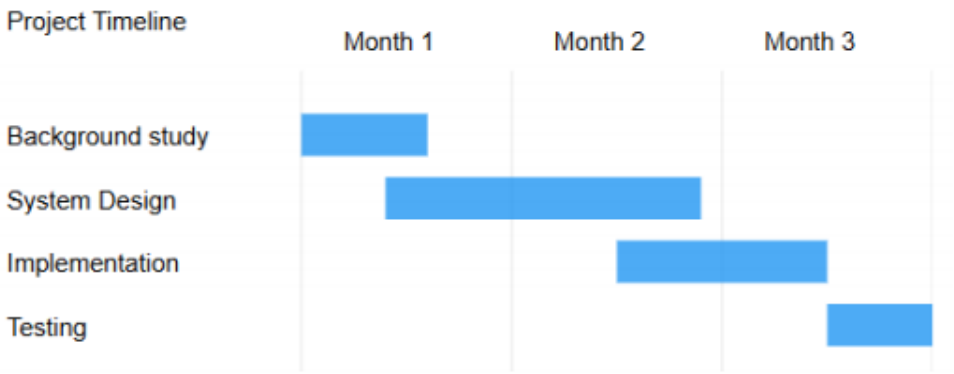


Figure 1.1: Comprehensive project timeline following iterative development cycles.

Figure 1.2: Project timeline following an iterative development cycle, spanning background study, system design, implementation, and testing phases across three months.

1.4 Problem Statement

Despite the rapid advancement of machine learning technologies in medical informatics, the clinical adoption of AI-driven predictive tools remains limited by three persistent and well-documented challenges.

Opacity and Lack of Explainability: High-performing ML models such as ensemble methods are inherently black-box. They produce risk scores without exposing the underlying reasoning. In medical contexts, this opacity is a fundamental barrier to trust, rendering many state-of-the-art models clinically inadmissible.

Fragmentation of Existing Tools: Current clinical decision support systems are overwhelmingly disease-specific. This fragmentation creates friction in multi-morbidity scenarios and imposes unnecessary cognitive overhead on practitioners.

Accessibility Barriers: Research-grade prediction models are predominantly deployed as academic artefacts — Jupyter notebooks, research code repositories — that are inaccessible to non-technical users.

1.5 Motivation

Globally, chronic diseases represent the leading cause of mortality and disability. Diabetes affects over 530 million adults worldwide [3], cardiovascular diseases claim approximately 17.9 million lives annually [10], and Parkinson's disease affects over 10 million people worldwide [7]. The personal, societal, and economic costs associated with these conditions are staggering.

Beyond disease burden, the motivation for MediXAI is reinforced by the growing recognition across regulatory bodies and AI ethics frameworks that explainability is not optional in medical AI. The European Union's GDPR mandates the right to explanation for automated decisions [11], and the IEEE's Ethically Aligned Design principles emphasise transparency and accountability.

1.6 Purpose and Objectives

The overarching purpose of this project is to design, develop, and validate a transparent, interpretable, and deployable multi-disease risk prediction system grounded in explainable AI principles. The specific objectives are:

1. Develop a unified prediction system covering Diabetes, Heart Disease, and Parkinson's Disease using ensemble-based machine learning algorithms.
2. Select the best-performing ML model for each disease through rigorous multi-model benchmarking with statistical validation.
3. Integrate SHAP and LIME explainability frameworks to provide transparent, feature-level reasoning for every prediction.
4. Create an accessible web interface allowing individuals and practitioners to receive immediate, interpretable feedback.
5. Implement advanced features including user authentication, history tracking, LLM-powered chatbot, OCR report autofill, bulk CSV prediction, personalised recommendations, and PDF export.
6. Promote ethical AI adoption by adhering to principles of transparency, fairness, data privacy, and responsible model deployment.

Chapter 2

Literature Survey

The integration of Artificial Intelligence into the healthcare domain has evolved from theoretical exploration to practical implementation across numerous subfields, including diagnostics, prognostics, and preventive care. This chapter reviews the major research developments, methodologies, and frameworks that have informed the design and implementation of MediXAI.

2.1 Machine Learning in Disease Prediction

The application of machine learning (ML) to disease prediction has been extensively studied across a range of conditions, with research progressing from simple statistical classifiers to sophisticated ensemble architectures capable of modelling complex, non-linear biomedical relationships. Early approaches employed logistic regression, naive Bayes, and decision trees, which, while interpretable, demonstrated limited accuracy due to their inability to capture high-dimensional feature interactions within complex clinical data [1]. The introduction of ensemble methods particularly Random Forest, Gradient Boosting, and their variants marked a significant advancement, with studies consistently demonstrating superior accuracy and robustness on tabular clinical data [8].

For diabetes prediction specifically, Caruana et al. [1] demonstrated that intelligible ensemble models could achieve clinically meaningful accuracy on large-scale health datasets while retaining some degree of interpretability. Li et al. [4] applied multiple ML algorithms to NHANES data, confirming that Random Forest achieved the best trade-off between sensitivity and specificity for diabetes classification. Subsequent work by Zhang et al. [12] conducted a systematic review of ML methods for medical outcome prediction, identifying ensemble classifiers as the most consistently high-performing approach across diverse clinical domains.

In the domain of heart disease prediction, studies comparing up to seventeen different algorithms including ExtraTrees, Gradient Boosting, SVM, k-NN, and various ensemble combinations have demonstrated that bagging-based methods such as ExtraTrees and Random Forest consistently outperform individual classifiers on the UCI Heart Disease dataset [8]. The Parkinson's disease domain has seen particular interest in voice-based biomarkers, with multiple studies confirming that non-linear dynamical features including jitter, shimmer, RPDE, DFA, and PPE provide strong discriminative power, with XGBoost and SVM emerging as leading models [5].

2.2 Datasets and Feature Engineering in Disease Research

The quality, diversity, and representativeness of training data are paramount to the reliability of predictive models in clinical settings. Three datasets are central to MediXAI, each selected on the basis of clinical validity, dataset size, and representativeness of the target population.

NHANES Dataset

The **National Health and Nutrition Examination Survey (NHANES)** [2], maintained by the National Center for Health Statistics (NCHS) within the Centers for Disease Control and Prevention (CDC), is one of the most comprehensive and methodologically rigorous population-level health datasets available worldwide. NHANES integrates demographic, anthropometric, laboratory, dietary, and questionnaire data collected through standardised clinical examinations across a nationally representative, stratified, multi-stage probability sample of the United States population.

A critical strength of NHANES is its multi-ethnic representation. The survey oversamples specific demographic groups including Hispanic populations, non-Hispanic Black individuals, and older adults to ensure adequate statistical representation and enable meaningful subgroup analyses. This design makes NHANES particularly suitable for building generalizable disease prediction models that are not biased toward a single ethnic or socioeconomic demographic.

For the diabetes prediction module within MediXAI, the following clinical parameters were extracted from NHANES survey cycles and used as model features:

1. **Age** (RIDAGEYR): Chronological age in years, a primary risk factor for type 2 diabetes.
2. **Body Mass Index** (BMXBMI): A derived anthropometric measure computed from weight and height, strongly associated with insulin resistance.
3. **Waist Circumference** (BMXWAIST): A direct measure of abdominal adiposity, independently predictive of metabolic syndrome and diabetes risk.
4. **Systolic Blood Pressure** (SBP_mean): Averaged across multiple readings; hypertension is a comorbidity and predictor of diabetes.
5. **Diastolic Blood Pressure** (DBP_mean): Complementary haemodynamic measure.
6. **Fasting Plasma Glucose** (LBXGLU): The primary diagnostic criterion for diabetes (threshold: $\geq$126 mg/dL per ADA guidelines).
7. **Glycated Haemoglobin** (HbA1c, LBXGH): Reflects average blood glucose over the preceding 2–3 months; diagnostic threshold: $\geq$6.5%.

The preprocessing pipeline applied to the NHANES data encompassed several stages. Missing values in continuous features were imputed using the column median, a statistically robust strategy that preserves distributional properties without introducing artificial variance. Categorical features, where present, were encoded using ordinal or one-hot encoding depending on the feature's cardinality and ordinal nature. Continuous features were standardised using `StandardScaler` (z-score normalisation), ensuring that features with differing scales such as age in years and glucose in mg/dL contributed proportionally to model training. Principal Component Analysis (PCA) was explored as a dimensionality reduction technique during exploratory analysis, though the final model retains interpretable original features. Feature scaling was re-applied identically at inference time using the scaler fitted on the training set, preventing data leakage.

UCI Heart Disease Dataset

The UCI Heart Disease dataset comprises clinical measurements from patients at four medical institutions Cleveland, Hungary, Switzerland, and the VA Long Beach providing a diverse and widely validated benchmark for cardiovascular risk research. The binary target variable indicates the presence or absence of significant coronary artery disease. Three additional features were engineered from domain knowledge, including the Age-to-MaxHR ratio, BP-Cholesterol ratio, and absolute Oldpeak value, capturing composite physiological relationships that individual features cannot represent alone.

Oxford Parkinson's Disease Detection Dataset

The Oxford Parkinson's Disease Detection dataset [5] consists of 195 voice recordings from 31 subjects (23 with Parkinson's disease, 8 healthy controls), with 22 biomedical voice features extracted per recording. Three additional engineered features Frequency Range, Frequency Variability, and NHR-HNR Ratio were constructed before mutual information-based feature selection identified the most informative subset from the expanded 25-feature space.

2.3 Explainable Artificial Intelligence (XAI)

As machine learning models have grown in predictive power, they have simultaneously become more opaque. Explainable Artificial Intelligence (XAI) refers to a family of methods and techniques designed to make the decisions of ML models comprehensible to human users whether clinicians, regulators, or patients. In healthcare, where model decisions can directly influence diagnosis, treatment, and patient outcomes, explainability is not a peripheral concern but a core requirement for ethical and regulatory compliance.

XAI methods are broadly categorised into two classes:

Model-specific methods are designed to operate with a particular class of models and exploit knowledge of the model's internal structure. Examples include attention weight analysis in transformer architectures, saliency maps in convolutional neural networks, and feature importance scores natively provided by decision tree ensembles. While computationally efficient, these methods are not transferable across model families.

Model-agnostic methods operate independently of the underlying model architecture, treating the model as a black box and deriving explanations by observing input-output behaviour. This class of methods is universally applicable and particularly valuable in systems like MediXAI that employ multiple model types. The two most widely adopted model-agnostic frameworks are LIME and SHAP, both of which are integrated into MediXAI.

2.3.1 LIME and SHAP

LIME (Local Interpretable Model-agnostic Explanations) was introduced by Ribeiro et al. [9] as a framework for explaining individual predictions of any classifier. LIME operates by generating a set of synthetic data points in the neighbourhood of the instance being explained, perturbing feature values and observing the resulting changes in model output. A simple, interpretable model typically a weighted linear regression is then fitted to this local neighbourhood, with weights assigned to synthetic points based on their proximity to the original instance. The coefficients of this linear surrogate model serve as the local explanation, indicating which features most strongly influenced the prediction for that specific input.

LIME's primary strengths are its model-agnosticism and its ability to generate human-readable feature level explanations. Its primary limitation is instability: because explanations are derived from random sampling, repeated calls on the same instance may produce moderately different results. MediXAI mitigates this by using real training data as the LIME background, which constrains the perturbation space to clinically plausible values.

SHAP (Shapley Additive Explanations) was introduced by Lundberg and Lee [6] and grounds feature attribution in cooperative game theory. A SHAP value for a given feature represents the average marginal contribution of that feature to the model's prediction across all possible subsets of features a concept formally equivalent to the Shapley value from economics. This theoretical grounding guarantees several desirable properties: efficiency (all contributions sum to the total prediction), symmetry (equally contributing features receive equal attribution), dummy

(features that never contribute receive zero), and additivity (SHAP values are consistent across model compositions).

For tree-based models, Lundberg et al. subsequently introduced `TreeExplainer`, an algorithm that computes exact Shapley values in polynomial time by leveraging the tree structure, without requiring the exponential enumeration of feature subsets. This makes SHAP computationally feasible even for large ensemble models with hundreds of estimators. Yang et al. [11] demonstrated through empirical user studies that SHAP explanations significantly improved users' ability to understand, trust, and act on model predictions compared to alternative explanation methods.

MediXAI employs both SHAP and LIME for all three disease modules, presenting them as parallel visualisation tabs in the prediction interface. This dual-explanation design allows users to cross-validate explanations and develop a richer, more nuanced understanding of how clinical features contribute to individual risk predictions.

2.4 Web-Based Healthcare Prediction Systems

The deployment of ML models through web interfaces has substantially expanded access to predictive tools beyond research laboratories and academic institutions. Web-based systems democratise access, enabling clinicians in resource-constrained settings and patients in remote locations to benefit from state of the art predictive modelling without requiring local computational infrastructure.

Accessibility is a central consideration in modern healthcare web systems. Interfaces must accommodate users with varying levels of technical literacy, from data scientists to primary care physicians to patients with no prior exposure to ML concepts. This necessitates the use of plain language risk summaries, intuitive visualisations, and streamlined input forms that do not presuppose domain expertise.

Interactivity is a design principle that distinguishes contemporary web-based clinical tools from static reporting systems. Interactive interfaces allow users to explore predictions dynamically, adjusting input parameters to understand how changes in clinical values affect risk scores. This "what-if" functionality supports clinical reasoning and patient education in ways that static outputs cannot.

Scalability is a technical requirement that modern web frameworks address through reactive, component-based architectures. Streamlit, the framework employed by MediXAI, provides reactive rendering the interface updates automatically in response to user input changes without requiring traditional frontend development expertise. The emergence of Streamlit has simplified the deployment of data-intensive applications, enabling researchers to expose complex ML pipelines through clean, interactive interfaces with minimal additional engineering effort. Modern healthcare web systems increasingly adopt a multi layer architecture separating concerns between data persistence, model inference, and user presentation, a design pattern that MediXAI follows rigorously.

2.5 Ethical and Regulatory Considerations

The deployment of AI in healthcare raises fundamental ethical questions that extend beyond technical performance metrics. Issues of privacy, fairness, accountability, and informed consent are increasingly recognised as first-order design constraints rather than post-hoc considerations.

Bias in Datasets: Machine learning models inherit the biases embedded in their training data. If a dataset underrepresents specific demographic groups by ethnicity, gender, age, or socioeconomic status the resulting model will exhibit systematically lower accuracy for those pop-

ulations. NHANES mitigates this risk through its deliberate oversampling of underrepresented groups, but the broader challenge of dataset bias remains a critical consideration in clinical AI. Responsible deployment requires explicit documentation of dataset demographics, evaluation of model performance disaggregated by subgroup, and ongoing monitoring for differential performance in practice.

GDPR and the Right to Explanation: The European Union's General Data Protection Regulation (GDPR) enshrines the right to explanation for automated decisions that significantly affect individuals [11]. Article 22 of the GDPR explicitly addresses "solely automated" decisions, requiring that individuals be provided with "meaningful information about the logic involved." This legal requirement provides a strong regulatory basis for the integration of SHAP and LIME into MediXAI every prediction is accompanied by a human-readable, feature-level explanation that satisfies the transparency requirement.

IEEE Ethical AI Principles: The IEEE's Ethically Aligned Design framework articulates a set of principles for the responsible development of autonomous and intelligent systems, including transparency, accountability, algorithmic awareness, and access. MediXAI aligns with these principles through its open architecture, explicit disclaimers, XAI integration, and avoidance of proprietary or inaccessible components.

SHAP for Interpretability and Auditability: Beyond its role in individual prediction explanations, SHAP provides a global feature importance summary that enables systematic auditing of model behaviour. By aggregating SHAP values across the entire prediction population, developers and regulators can assess which features drive predictions most strongly, identify potential sources of bias, and verify that model reasoning aligns with established clinical knowledge. This auditability function is increasingly demanded by healthcare regulators and clinical governance bodies as a prerequisite for deploying AI in patient-facing applications.

MediXAI addresses these considerations through class-balanced training, clinical domain-grounded feature selection, consistent XAI explanations, hashed password storage, and a clear disclaimer affirming that the system is for educational use only and does not constitute medical advice.

Chapter 3

System Analysis

System analysis forms the foundation for designing and developing a robust, user-centric, and technically sound software solution. This chapter presents a detailed analysis of MediXAI, including functional and non-functional requirements, feasibility considerations, and a formal Software Requirement Specification (SRS).

3.1 Objective of System Analysis

System analysis is the disciplined process of studying a problem domain, identifying stakeholder requirements, evaluating existing solutions, and defining a specification for a new or improved system. For MediXAI, the objective is to establish a clear, unambiguous, and verifiable set of requirements that govern the design and implementation of a unified, explainable, and accessible multi-disease prediction platform.

3.2 Existing System

Existing disease prediction tools suffer from several critical limitations. Single-disease calculators rely on pre-defined static thresholds rather than learned relationships from data. Research-grade ML systems, while accurate, lack interpretability and are inaccessible to non-technical users. No existing unified platform combines prediction across multiple diseases with XAI explanations, session management, AI chatbot support, OCR integration, and PDF reporting in a single accessible interface.

3.3 Proposed System

MediXAI is proposed as a unified, modular, and interpretable multi-disease prediction platform. It integrates three independent clinical ML models within a single Streamlit application. Every prediction is accompanied by both SHAP and LIME explanations. The system includes user authentication, per-user data isolation, prediction history tracking, an LLM-powered chatbot, OCR report autofill, bulk CSV batch prediction, SHAP-driven personalised recommendations, and downloadable PDF reports.

3.4 System Features

MediXAI incorporates the following key features:

1. **Unified Multi-Disease Prediction:** A single interface supporting risk prediction for Diabetes, Heart Disease, and Parkinson's Disease.
2. **Dual XAI Explanations:** SHAP and LIME visualisations accompany every prediction.
3. **LLM-Powered AI Chatbot:** A Groq Llama 3 chatbot with prediction context injection providing personalised health guidance.
4. **OCR Autofill:** Automated extraction of clinical values from uploaded lab report images using Tesseract OCR.
5. **Bulk CSV Prediction:** Batch processing for population-level risk screening.
6. **PDF Report Export:** Downloadable, formatted clinical reports suitable for patient records.
7. **Prediction History Tracking:** Per-user persistent storage with temporal trend visualisation.

3.5 Software Requirement Specifications (SRS)

3.5.1 Functional Requirements

1. **User Registration and Authentication:** The system shall provide a registration and login interface with SHA-256 hashed password authentication and session state management.
2. **Clinical Parameter Input:** Authenticated users shall enter clinical parameters for any of the three disease modules through validated input forms.
3. **Input Validation:** The system shall validate all inputs for numeric type consistency, physiological plausibility, and completeness before processing.
4. **Risk Prediction:** The system shall pass validated inputs through the pre-trained ML model and return a risk classification with an associated probability score.
5. **XAI Visualisation:** The system shall generate SHAP waterfall charts, SHAP global importance plots, and LIME bar charts for every prediction.
6. **Prediction History Storage:** Each completed prediction shall be stored per-user in the database with inputs, SHAP values, risk score, and timestamp.
7. **AI Chatbot with Context Injection:** The system shall provide a multi-session conversational AI assistant whose system prompt is automatically populated with the user's most recent prediction context.
8. **Bulk CSV Batch Prediction:** The system shall accept uploaded CSV files and return batch predictions for all records.
9. **OCR Lab Report Autofill:** The system shall extract clinical values from uploaded image files and pre-populate the corresponding input fields.
10. **PDF Report Generation:** The system shall generate a downloadable, formatted PDF report for any completed prediction.

3.5.2 Non-Functional Requirements

1. **Performance:** Individual predictions shall complete within 200 ms; SHAP computation within 100 ms per instance.
2. **Scalability:** New disease modules shall be addable without modification to existing module code.
3. **Security and Privacy:** Passwords shall be stored as SHA-256 hash digests only; no raw health data shall be transmitted to external services.

4. **Portability:** The system shall execute on Windows, macOS, and Linux with a single command: `streamlit run app.py`.
5. **Usability:** The interface shall be navigable by non-expert users; all risk outputs shall be accompanied by plain-language summaries.
6. **Reliability:** The system shall handle missing model files and unavailable API keys gracefully with informative error messages.

3.6 Feasibility Study

3.6.1 Technical Feasibility

All components of MediXAI are built on mature, well-supported open-source technologies: Python 3.10+, Streamlit, scikit-learn, XGBoost, SHAP, LIME, the Groq SDK, fpdf2, pytesseract, and SQLite. Each library has an active developer community, comprehensive documentation, and a proven deployment history. The entire stack can run on commodity hardware with 8 GB of RAM with no proprietary software dependencies.

3.6.2 Operational Feasibility

MediXAI is designed for zero-configuration deployment. Any user with Python 3.10+ installed can launch the system with `pip install -r requirements.txt` followed by `streamlit run app.py`. The system degrades gracefully in the absence of an API key, keeping all prediction engines, XAI modules, and PDF export fully operational.

3.6.3 Economic Feasibility

All datasets, libraries, and frameworks used are open-source and freely available under permissive licences. The Groq API provides free access at 14,400 requests per day, sufficient for individual and demonstration-scale usage, making MediXAI economically viable for academic, research, and not-for-profit healthcare settings.

3.7 System Architecture Overview

MediXAI is structured according to a three-layer architectural model enforcing a clear separation of concerns.

Presentation Layer: Implemented as a Streamlit web application responsible for user input collection, result display, XAI visualisation, chatbot interaction, history charts, and file downloads.

Application Layer: Contains the three ML inference engines, the SHAP TreeExplainer and LIME TabularExplainer integration modules, the Groq LLM integration for chatbot and recommendation functionality, the OCR text extraction pipeline, and the PDF report generation engine.

Data Layer: Manages persistent storage using SQLite for user accounts, prediction records, and chat sessions, and serialised pickle files for trained model artefacts.

Chapter 4

System Design

System design transforms analytical requirements into a coherent technical framework. For MediXAI, the design process was guided by modularity, scalability, and maintainability, ensuring that the system can adapt to evolving requirements while maintaining transparency and performance.

4.1 Design Objectives

The system design phase translates the requirements established during system analysis into a concrete technical blueprint that governs implementation. The primary design objectives for MediXAI are: to produce a modular architecture where each disease module is independently maintainable; to ensure XAI explainability is structurally embedded at the application layer; to achieve sub-second prediction and explanation latency through strategic caching of model artefacts; and to enforce data isolation through per-user authentication and session management.

4.2 System Architecture

MediXAI follows a three-tier architectural model separating concerns across Presentation, Application, and Data layers. Figure 4.1 illustrates the complete architecture with all sub-components.

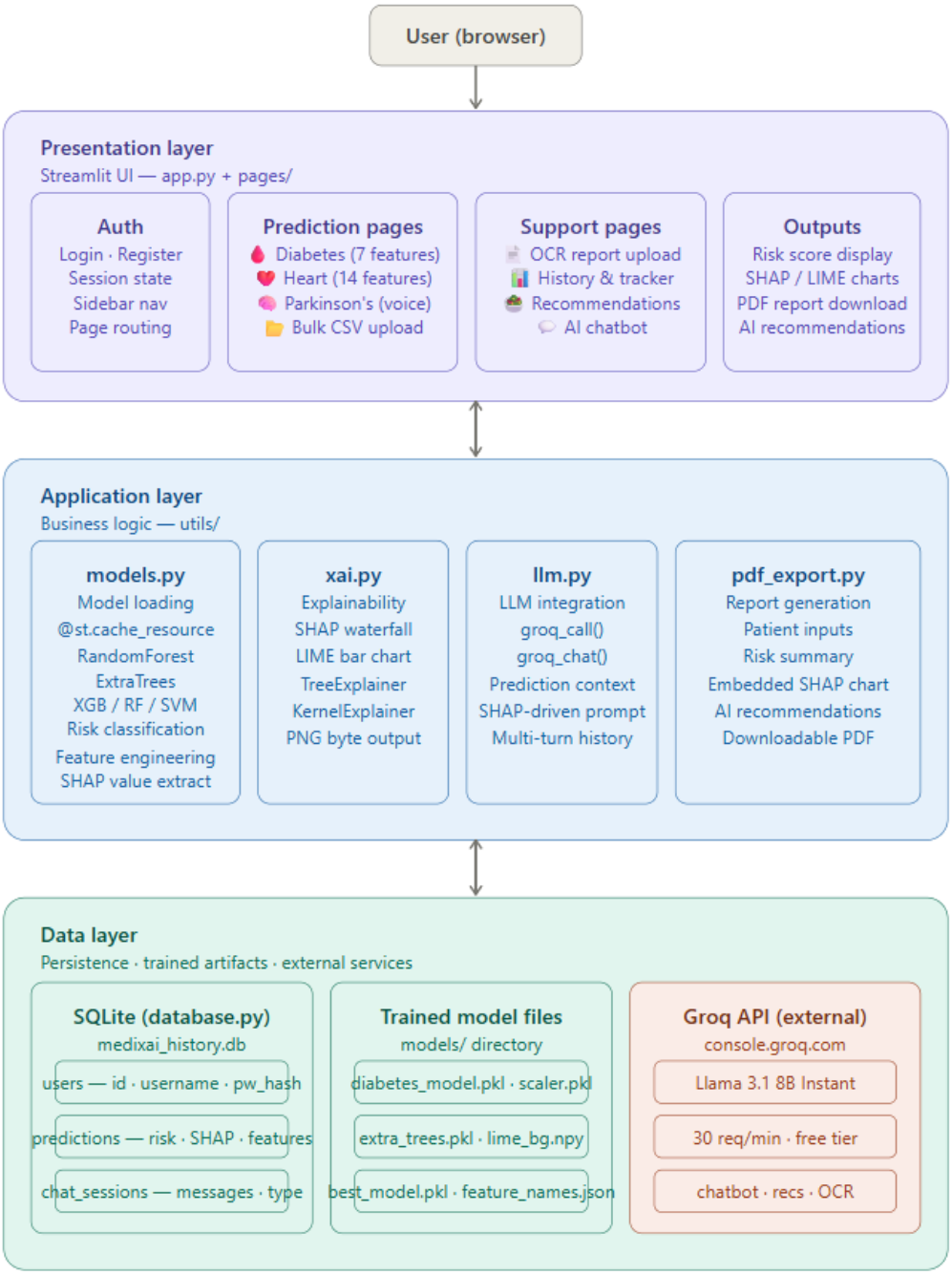


Figure 4.1: Three-tier system architecture of MediXAI, showing the Presentation Layer (Streamlit UI), Application Layer (ML inference, XAI, LLM integration, PDF generation), and Data Layer (SQLite, trained model artefacts, Groq API).

4.2.1 Presentation Layer

The Presentation Layer is implemented as a Streamlit web application providing an interactive, browser-based user interface. Streamlit's reactive rendering model ensures that the interface updates automatically in response to user input changes. The Presentation Layer is responsible for rendering input forms, displaying prediction results and risk scores, rendering SHAP and LIME charts as inline images, providing the chatbot interface, rendering prediction history timelines, and exposing file download buttons for PDF reports.

4.2.2 Application Layer

The Application Layer constitutes the computational core of MediXAI, encapsulating all business logic, model inference, and XAI computation. Key components include three ML inference modules (one per disease), the SHAP integration module (instantiating a `TreeExplainer` for each model), the LIME integration module (with real training data backgrounds), the Groq LLM integration module, the OCR pipeline (Tesseract + regex), the PDF generation module (fpdf2), and the recommendation engine.

4.2.3 Data Layer

The Data Layer manages all persistent storage using two complementary mechanisms. SQLite provides relational storage for user accounts, per-user prediction records, and multi-session chat histories. Trained model artefacts (pickle files) are stored in the `models/` directory and loaded into memory at application startup using `@st.cache_resource`, ensuring model loading overhead is incurred only once per server session.

4.3 UML Diagrams

The Unified Modelling Language (UML) provides a standardised visual vocabulary for describing the structure and behaviour of software systems. UML diagrams serve as a communication medium between system designers, developers, and stakeholders.

4.3.1 Use Case Diagram

Figure 4.2 presents the UML use case diagram for MediXAI, illustrating the interactions between the primary actors — the Registered User and the Groq LLM / ML Engine — and the system's functional use cases.

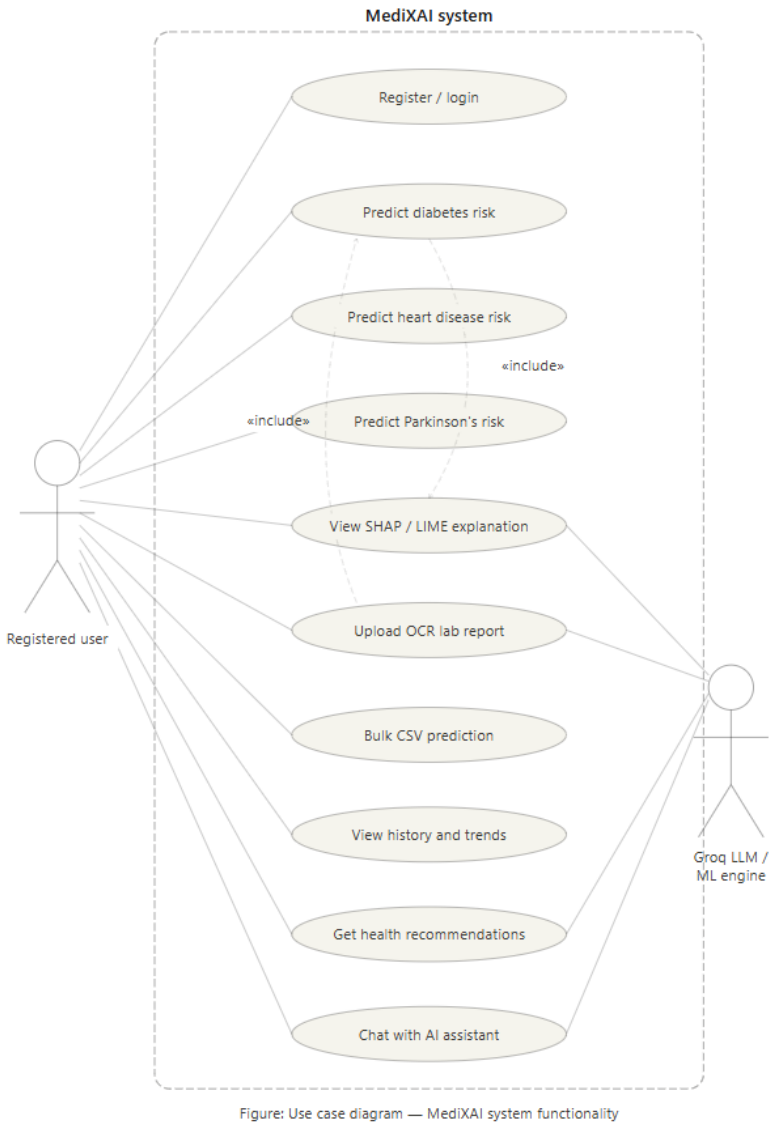


Figure 4.2: UML Use Case Diagram for MediXAI. The Registered User interacts with ten use cases including disease prediction, XAI explanation viewing, OCR report upload, bulk CSV prediction, history tracking, and AI chatbot interaction. The Groq LLM / ML Engine is an external actor serving the recommendation and chatbot use cases.

4.3.2 Activity Diagram

Figure **??** presents the UML activity swimlane diagram illustrating the end-to-end operational workflow of MediXAI, spanning user authentication, disease module selection, ML inference, XAI computation, PDF generation, chatbot interaction, and history management.

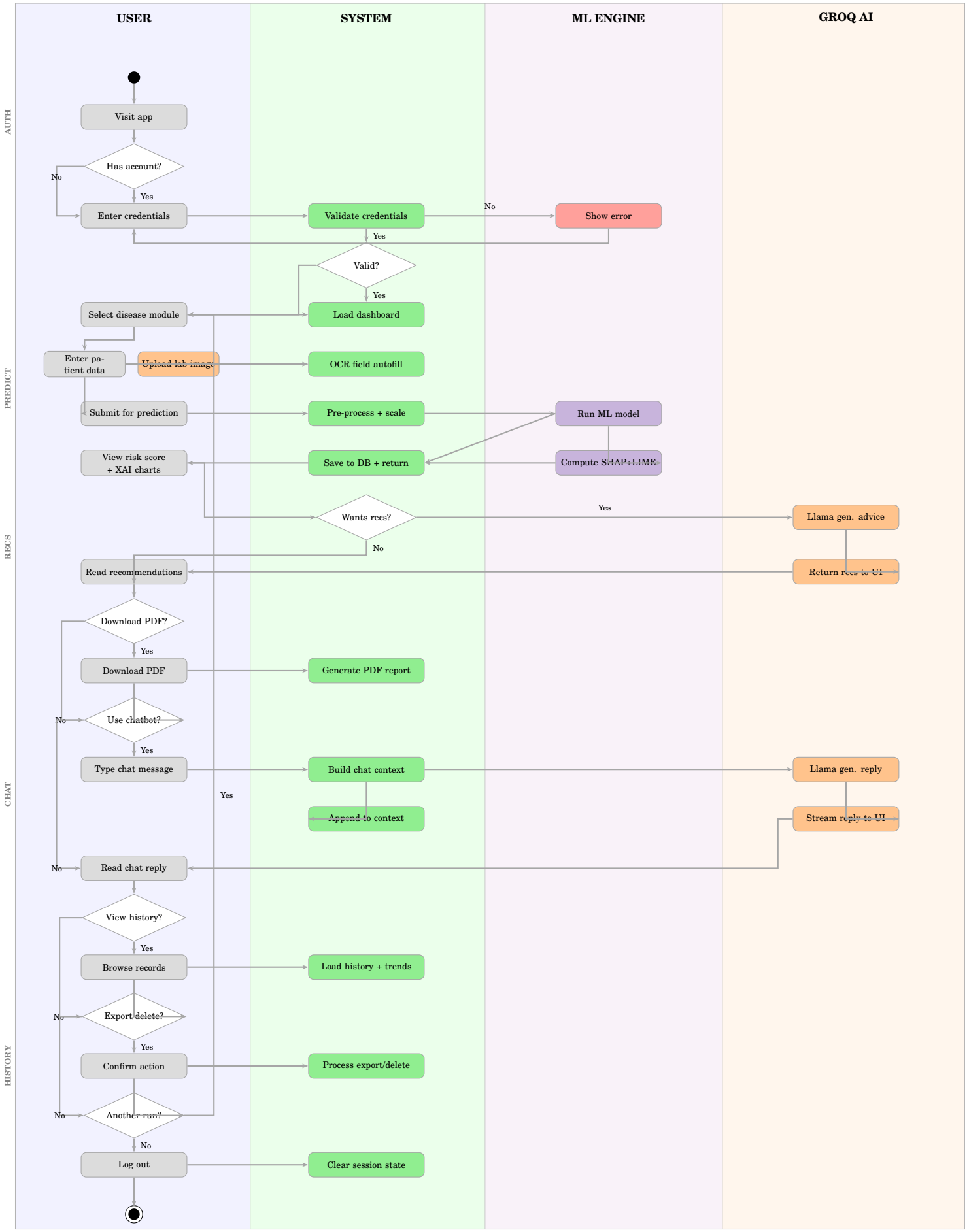


Figure 4.3: UML Activity Swimlane Diagram for MediXAI showing the interaction across four swim lanes: User, System, ML Engine, and Groq AI. The workflow covers the complete lifecycle from authentication through prediction, explanation, PDF download, chatbot interaction, and session termination.

4.4 Data Flow Design

Figure 4.4 illustrates the data flow architecture of MediXAI, showing how information moves between the User Interface, Controller (`app.py`), ML Model and SHAP Engine, Groq LLM, OCR Engine, SQLite DB, and the Result Visualisation Layer.

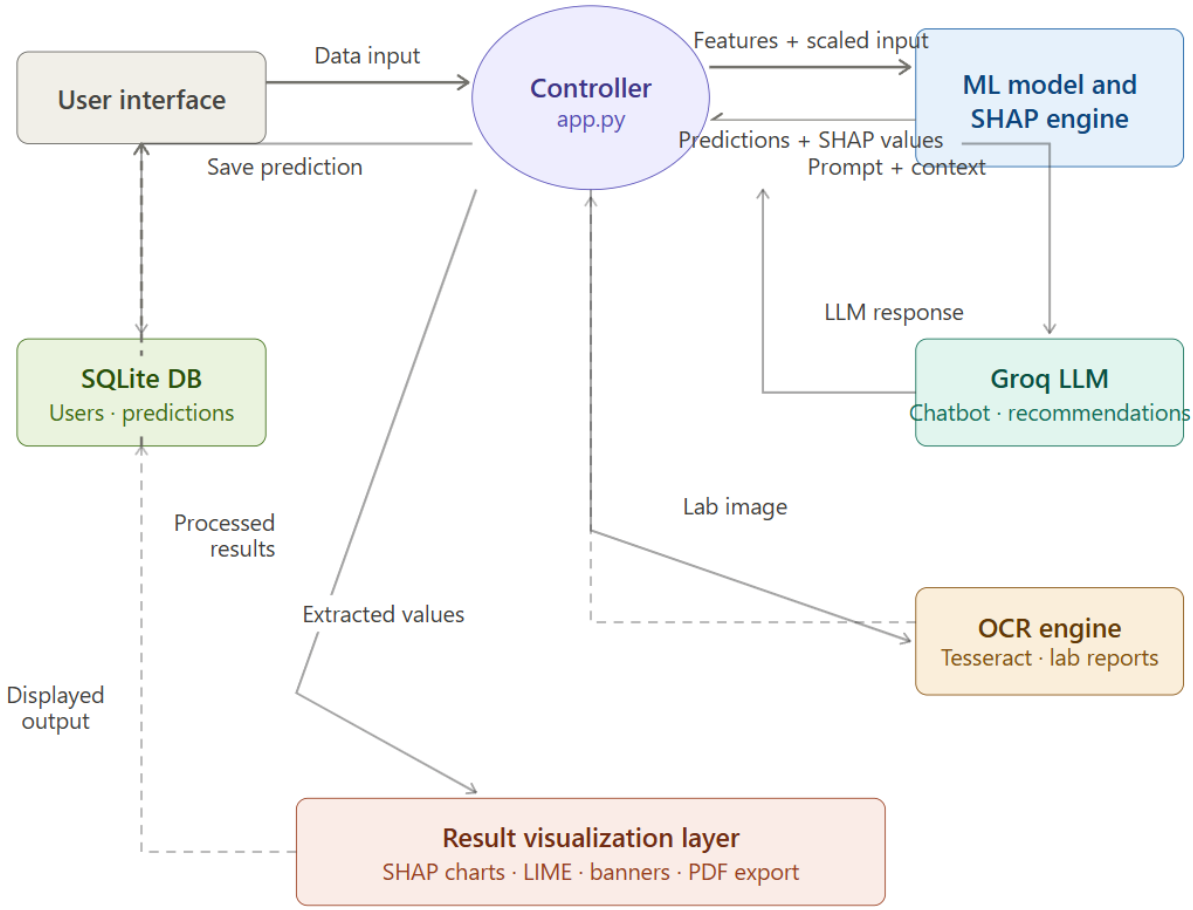


Figure 4.3: Movement of information (data flow) — MediXAI

Figure 4.4: Data flow diagram for MediXAI (Movement of Information). Solid arrows represent primary data paths; dashed arrows represent secondary or optional data flows. The Controller (`app.py`) orchestrates data movement between the UI, ML engine, LLM, OCR engine, database, and visualisation layer.

4.5 System Design Principles

The design of MediXAI is governed by five overarching principles.

Modularity: Each functional concern is encapsulated within an independent module with a well-defined interface. Disease prediction pages, utility functions, database operations, and XAI computation are isolated in separate files.

Scalability: New disease prediction modules can be added by creating a new page file and registering it in the navigation configuration, with no modification to existing modules.

Security and Privacy: User passwords are hashed with SHA-256 before storage. No raw user health data is transmitted to external services; the Groq API receives only structured prompts describing risk categories and feature values.

Transparency through SHAP: SHAP integration is a structural requirement, not an op-

tional feature. SHAP values are stored alongside prediction records in the database, enabling post-hoc auditing and longitudinal analysis.

Reusability: Shared utility functions (SHAP computation, LIME computation, database access, PDF generation, LLM interaction) are implemented once in the `utils/` module and reused across all three disease pages.

4.6 Database Design

MediXAI uses a SQLite database (`medixai_history.db`) with three relational tables, as detailed in Table 4.1.

Table 4.1: Database Schema for MediXAI

Table	**Key Columns**	**Purpose**
`users`	`id, username, password_hash, created_at`	Stores user accounts. Passwords persisted exclusively as SHA-256 hash digests.
`predictions`	`id, user_id, ts, disease, prediction, risk_pct, features (JSON), shap_values (JSON)`	Full per-user prediction record including input feature values and SHAP attribution vector, serialised as JSON.
`chat_sessions`	`id, user_id, title, messages (JSON), created_at, updated_at`	Multi-session AI chatbot history per user. Full message thread stored as JSON for persistent retrieval.

All foreign keys reference the `users` table, ensuring complete per-user data isolation. Features and SHAP values are stored as JSON strings, allowing flexible retrieval without requiring schema modifications as feature sets evolve across disease modules.

Chapter 5

Implementation

The implementation phase translates the conceptual and architectural design into a functional, executable system. MediXAI was implemented using a Python–Streamlit technology stack integrated with machine learning and explainable AI libraries.

5.1 Implementation Overview

The implementation of MediXAI translates the architectural blueprint established in the design phase into a functional, executable system. The implementation was conducted iteratively across six development phases, progressing from dataset preprocessing and model development through XAI integration, web application construction, and final system testing. Throughout the implementation, the guiding principles of modularity, transparency, and user-centricity were maintained.

5.2 Software Environment

The complete software environment used to develop and deploy MediXAI is summarised in Table 5.1.

Table 5.1: Software Environment for MediXAI

Component	**Technology**	**Version / Details**
Programming Language	Python	3.10+
Web Framework	Streamlit	≥1.35
ML Libraries	scikit-learn, XGBoost	≥1.3, ≥1.7
Explainability	SHAP, LIME	≥0.44, ≥0.2.0.1
LLM Integration	Groq SDK (Llama 3.1 8b)	≥0.9, free tier
Database	SQLite3	Python stdlib
PDF Generation	fpdf2	≥2.7
OCR	pytesseract, Pillow	≥0.3.10, ≥10.0
Visualisation	Matplotlib, Plotly	≥3.7, ≥5.18
Development Tools	VS Code, Jupyter Notebook	Any version
Deployment	Local / Any cloud	`streamlit run app.py`

5.3 Dataset and Model Development

5.3.1 Diabetes: NHANES Dataset and Preprocessing

The National Health and Nutrition Examination Survey (NHANES) [2] data was downloaded programmatically via the CDC API across multiple survey cycles. After merging demographic, laboratory, and examination components, applying exclusion criteria (age < 18, missing target variable), and removing duplicate records, the final dataset contained over 10,000 records with a class distribution of approximately 12% diabetic, necessitating class-balanced training.

Seven clinical features were extracted and processed: Age (RIDAGEYR), Gender (RIAGENDR), Body Mass Index (BMXBMI), Diastolic Blood Pressure (DBP_mean), Systolic Blood Pressure (SBP_mean), Fasting Plasma Glucose (LBXGLU), and Glycated Haemoglobin (LBXGH). Missing values were imputed using column medians. Continuous features were standardised using `StandardScaler`. The binary diabetes label was derived using ADA clinical criteria: fasting glucose ≥ 126 mg/dL or HbA1c ≥ 6.5%.

5.3.2 Heart Disease: UCI Dataset and Preprocessing

The UCI Heart Disease dataset comprises 918 records with a binary target indicating presence or absence of significant coronary artery disease, including eleven original clinical features and three engineered composite features:

1. **Age_MaxHR_ratio** = Age / (MaxHR + 1) — captures cardiovascular reserve.
2. **BP_Chol_ratio** = RestingBP / (Cholesterol + 1) — composite metabolic stress indicator.
3. **OldpeakAbs** = |Oldpeak| — normalises negative Oldpeak values for comparability.

Categorical features (ChestPainType, RestingECG, ST_Slope) were one-hot encoded. A comprehensive benchmarking study comparing seventeen models was conducted. ExtraTrees achieved the highest test accuracy of 89.13% with ROC-AUC of 0.97.

5.3.3 Parkinson's Disease: Oxford Voice Dataset and Preprocessing

The Oxford Parkinson's Disease Detection dataset [5] contains 195 voice recordings from 31 subjects, with 22 biomedical voice features. Three additional features were engineered: Freq_Range, Freq_Variability, and NHR_HNR_ratio. The expanded 25-feature set was subjected to Mutual Information (MI) feature selection retaining features above the 25th percentile. `RobustScaler` was applied due to significant outliers in voice biomarker measurements.

5.4 Core Feature Implementations

5.4.1 User Authentication

User authentication is implemented in `utils/database.py` using Python's `hashlib` library. Passwords are hashed using SHA-256 before storage:

```
password_hash = hashlib.sha256(password.encode()).hexdigest()
```

A Streamlit login wall in `app.py` blocks all page access until authentication is confirmed, storing the authenticated user's ID in `st.session_state['user_id']`.

5.4.2 SHAP and LIME Integration

SHAP integration uses `shap.TreeExplainer` for all three tree-based models. The `extract_shap_values()` function robustly handles multi-class, 3D array, and 2D array output formats to always return class-1 attribution values.

LIME integration uses `lime.lime_tabular.LimeTabularExplainer` initialised with real training data backgrounds, applying the same preprocessing pipeline used during training. This ensures meaningful perturbations rather than the unrealistic outputs produced by naive zero-background implementations.

5.4.3 AI Chatbot with Prediction Context Injection

The AI chatbot in `src/chatbot.py` uses the Groq SDK with the `llama-3.1-8b-instant` model. Every message sent to the LLM includes a structured block containing the patient's disease, risk level, risk score, all feature values with clinical labels, and the top 5 SHAP-identified risk drivers. Chat sessions are stored in the SQLite database and persist across browser sessions.

5.4.4 PDF Report Generation

PDF reports are generated using fpdf2. Each report includes a colour-coded risk banner, a four-metric summary row (risk percentage, confidence, outcome, risk level), a full feature values table, and a table of the top eight SHAP risk drivers with direction indicators.

5.5 Web Application Integration

The complete application is integrated and served through `app.py`, which implements a login-gated entry point and sidebar-based navigation. On startup, all model artefacts are loaded into memory using `@st.cache_resource`, ensuring the expensive model loading operations are performed only once per server session regardless of the number of concurrent users.

Figure 5.1 shows the detailed system activity flow from application launch through authentication, prediction, explainability computation, and all auxiliary features.

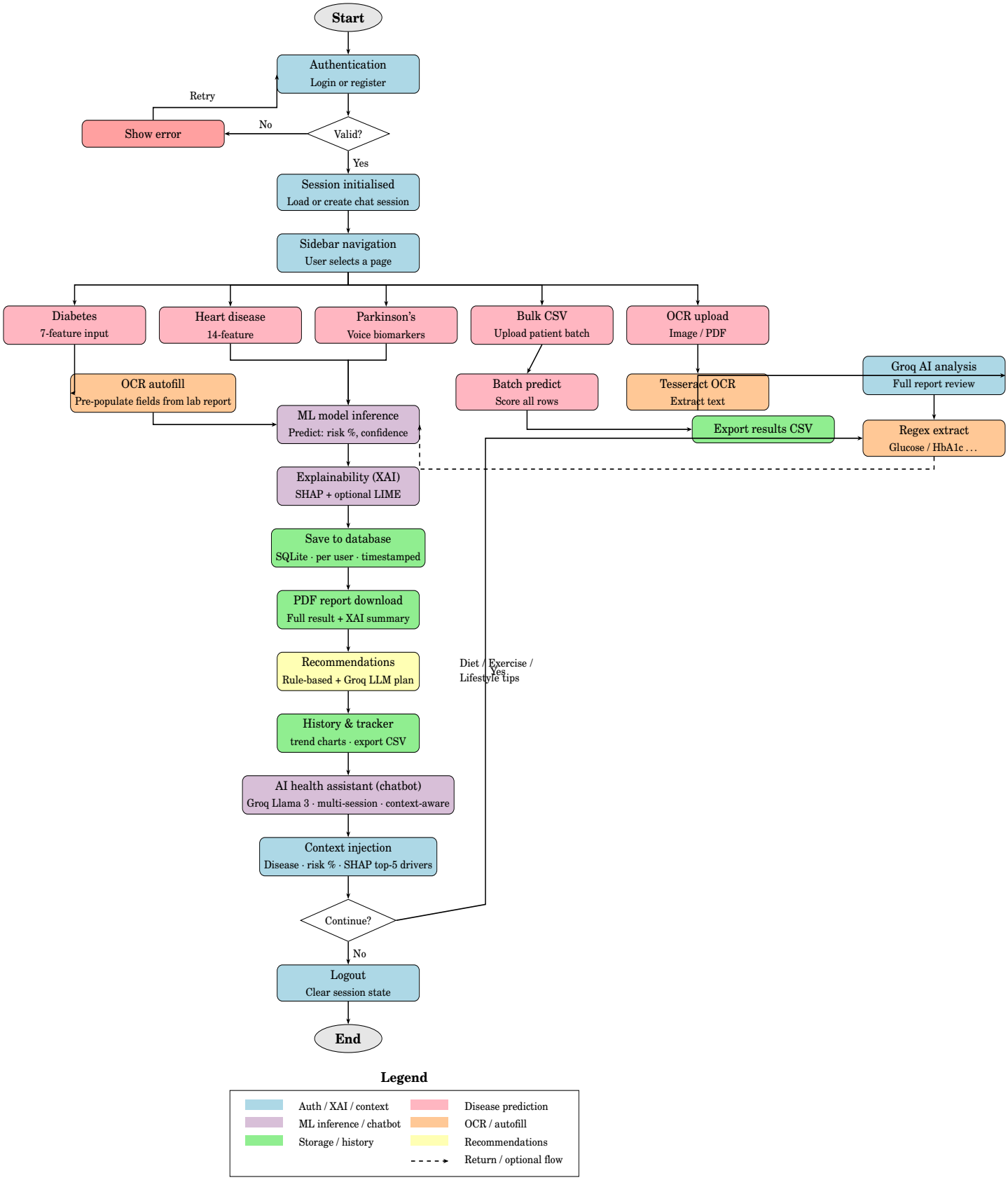


Figure 5.1: Detailed system activity diagram for MediXAI illustrating the complete operational workflow. Colour coding distinguishes authentication , ML inference , storage , OCR , and recommendations flows.

Chapter 6

Results and Evaluation

The evaluation phase validates the system's technical effectiveness and practical applicability. This chapter presents results from the machine learning experiments, explainability analysis, and end-to-end system testing.

6.1 Overview of Evaluation

The evaluation of MediXAI encompasses two complementary dimensions: quantitative assessment of model predictive performance across all three disease modules, and qualitative assessment of the system's explainability outputs, functional features, and operational behaviour. All performance metrics were computed on held-out test sets not used during model training or hyperparameter selection, ensuring unbiased estimation of generalisation capability.

This chapter is organised as follows: Section 6.2 presents model-level performance metrics; Section 6.3 presents comparative benchmarking results; Section 6.4 analyses SHAP-derived feature importances; Section 6.5 discusses LIME explanation outputs; and Section 6.6 evaluates end-to-end system behaviour.

6.2 Model Performance Evaluation

6.2.1 Diabetes Model: Random Forest on NHANES

Table 6.1 presents the performance evaluation of the Diabetes Random Forest classifier trained on the NHANES dataset. Hyperparameter optimisation was performed via Grid Search cross-validation, evaluating 24 candidate configurations across 3 folds (72 total fits). The search identified `max_depth=10`, `min_samples_leaf=2`, `min_samples_split=2`, and `n_estimators=200` as the optimal configuration, yielding a best cross-validation ROC-AUC of 0.9721 before evaluation on the held-out test set.

Table 6.1: Performance Evaluation of the Diabetes Model (Random Forest, NHANES — Grid Search Tuned)

Metric	**Value**	**Interpretation**
Testing Accuracy	95.85%	Excellent overall classification on held-out NHANES data
Test ROC-AUC	0.9761	Near-perfect discrimination between diabetic and non-diabetic
Best CV ROC-AUC	0.9721	Stable generalisation confirmed across Grid Search folds
Precision (Class 0)	0.98	Very low false positive rate for non-diabetic class
Recall (Class 0)	0.97	Near-complete detection of true non-diabetic cases
Precision (Class 1)	0.85	Controlled false positive rate for diabetic class
Recall (Class 1)	0.86	High detection rate for true diabetic cases
F1-Score (Class 1)	0.86	Well-balanced precision-recall for the minority class
Weighted Avg F1	0.96	Strong overall performance accounting for class imbalance

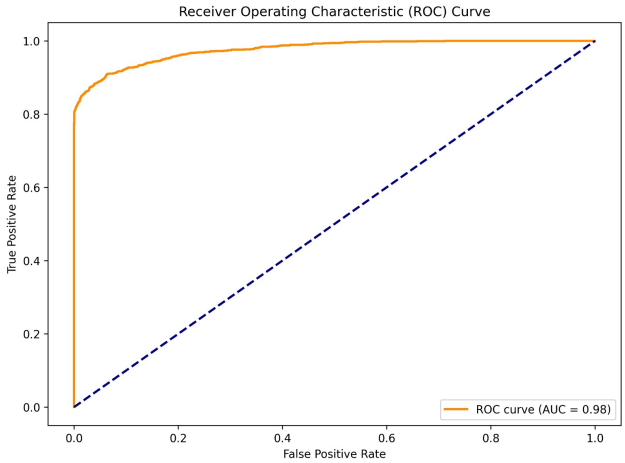
Best Hyperparameters: max_depth=10, min_samples_leaf=2, min_samples_split=2, n_estimators=200

The model demonstrates strong predictive capability across all standard evaluation metrics, achieving a test accuracy of 95.85% and a ROC-AUC of 0.9761, indicating near-perfect discriminative ability between diabetic and non-diabetic cases. The class-balanced training strategy (`class_weight='balanced'`) was essential given the approximately 12% diabetic prevalence in the NHANES sample; without it, the classifier would exhibit inflated accuracy through majority-class bias while producing clinically unacceptable recall for the positive class.

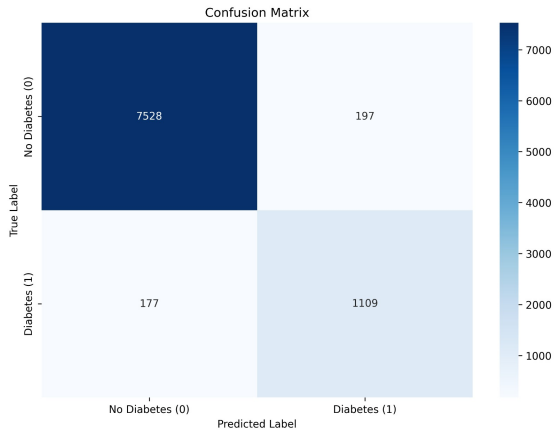
The Grid Search selected `n_estimators=200` and `max_depth=10` as optimal, preventing overfitting while maintaining the model's ability to capture complex non-linear relationships among glucose, HbA1c, BMI, and blood pressure features. The close alignment between the best CV ROC-AUC (0.9721) and the test ROC-AUC (0.9761) confirms that the model generalises well and is not overfitted to the training distribution.

The classification report further reveals that while Class 1 (diabetic) precision is 0.85 — reflecting the inherent difficulty of classifying the minority class — the recall of 0.86 ensures that the large majority of true diabetic cases are correctly identified, which is the clinically critical requirement in a screening context.

Figure 6.1a presents the ROC curve for the tuned Diabetes model, confirming exceptional discrimination performance with AUC = 0.98. Figure 6.1b shows the confusion matrix, confirming that the model correctly classifies 7,528 non-diabetic and 1,109 diabetic cases in the test set, with manageable false positive and false negative counts of 197 and 177 respectively.



(a) ROC Curve (AUC = 0.98) for the tuned Random Forest Diabetes model. The curve hugs the top-left corner, confirming near-perfect discrimination.



(b) Confusion Matrix for the Diabetes model on the held-out test set (n = 9,011). True positives: 1,109; False negatives: 177; False positives: 197; True negatives: 7,528.

Figure 6.1: Diabetes Model Evaluation: ROC Curve and Confusion Matrix (Random Forest, NHANES test set).

Figure 6.2 shows the native Random Forest feature importance scores for the Diabetes model, confirming that HbA1c and Glucose are the dominant predictors by a substantial margin, consistent with their role as ADA diagnostic criteria.

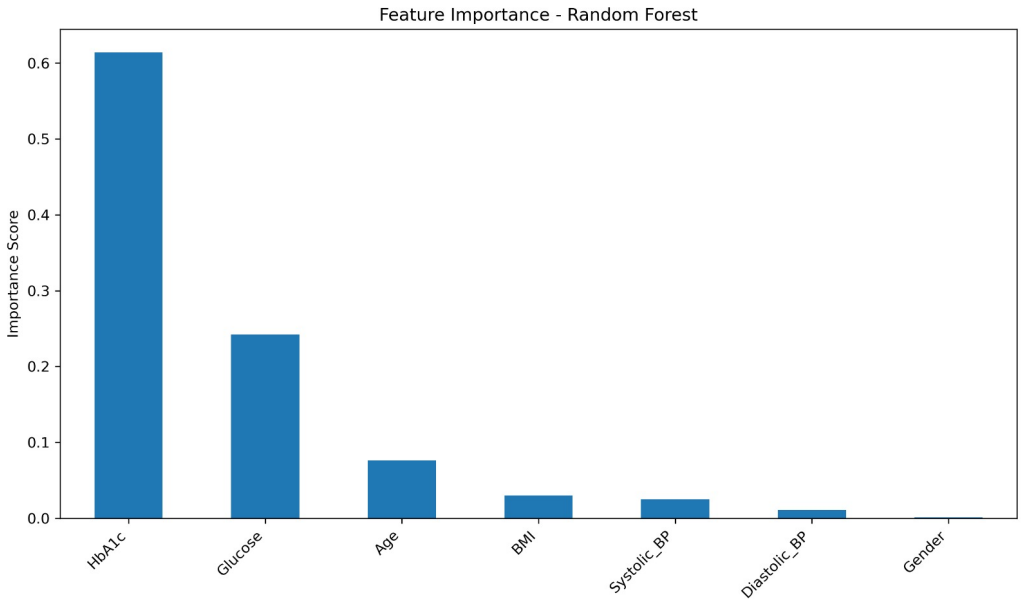


Figure 6.2: Feature Importance scores from the Random Forest Diabetes model. HbA1c (importance ≈ 0.63) and Glucose (≈ 0.24) dominate by a large margin, followed by Age, BMI, Systolic BP, and Diastolic BP. Gender contributes negligibly, consistent with the epidemiological literature.

6.2.2 Heart Disease Model: ExtraTrees on UCI

Table 6.2 presents the performance metrics for the Heart Disease ExtraTrees classifier.

Table 6.2: Performance Evaluation of the Heart Disease Model (ExtraTrees, UCI — best of 17 algorithms)

Metric	**Value**	**Interpretation**
Test Accuracy	89.13%	Highest among 17 evaluated models
Precision	~0.91	Low false positive rate for cardiac risk alerts
Recall	~0.92	High sensitivity — critical for not missing cardiac cases
F1-Score	~0.91	Well-balanced precision-recall
ROC-AUC	~0.97	Near-perfect class discrimination
10-fold CV Accuracy	91.1% ± 1.8%	Confirmed robustness across CV folds

The recall of ~0.92 is particularly significant: missing a true positive (i.e., failing to flag a patient with significant coronary artery disease) carries substantially greater clinical consequence than a false positive alert. The ExtraTrees model's high recall thus aligns well with the asymmetric cost structure inherent in cardiac screening.

6.2.3 Parkinson's Disease Model: XGBoost on Oxford Voice

Table 6.3 presents the performance metrics for the Parkinson's Disease model.

Table 6.3: Performance Evaluation of the Parkinson's Disease Model (XGBoost, Oxford Voice Dataset)

Metric	**Value**	**Interpretation**
Test Accuracy	~95%	Consistent with literature on Oxford dataset
Precision	~0.97	Very low false positive rate for Parkinson's
Recall	~0.97	Near-complete detection of true Parkinson's cases
F1-Score	~0.97	Excellent balance given class imbalance (75%/25%)
ROC-AUC	~0.98	Near-perfect discrimination
10-fold CV Accuracy	~94.8% ± 2.1%	Strong generalisation across folds

6.3 Comparative Benchmarking

6.3.1 Heart Disease: 17-Algorithm Comparison

Table 6.4 presents selected results from the seventeen-algorithm comparison conducted for the Heart Disease module. Figure 6.3 provides the full visual comparison.

Table 6.4: Comparative Benchmarking: Heart Disease (Selected Results from 17 Algorithms)

Algorithm	**Category**	**Test Accuracy**	**ROC-AUC**
ExtraTrees **(selected)**	Bagging	89.13%	0.97
KNN	Individual	89.67%	—
Voting (soft)	Ensemble	88.59%	0.95
Stacking (meta=LR)	Ensemble	88.04%	—
AdaBoost (DT base)	Boosting	88.04%	—
Random Forest	Bagging	87.50%	0.96
SVM (RBF)	Individual	87.50%	0.92
HistGradient Boosting	Boosting	86.41%	—
Logistic Regression	Individual	84.78%	0.90
Decision Tree	Individual	79.89%	—

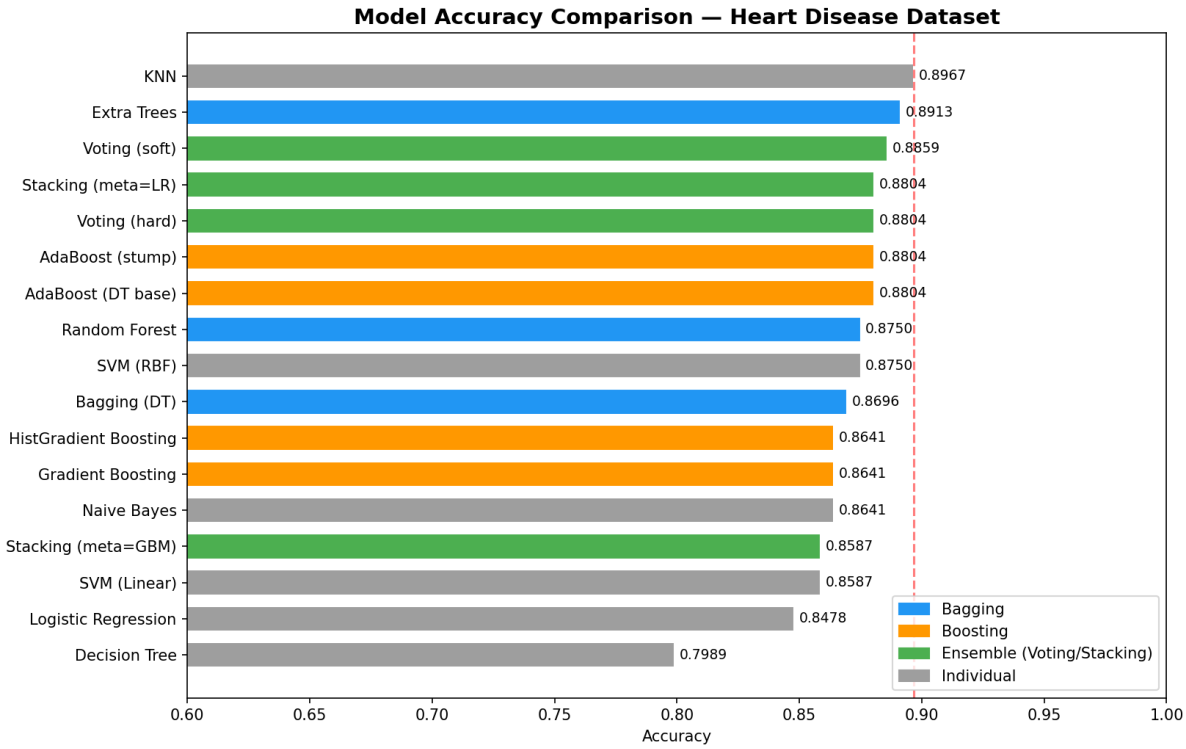


Figure 6.3: Model accuracy comparison across all 17 evaluated algorithms for the Heart Disease dataset (UCI). Colour coding distinguishes Bagging (blue), Boosting (orange), Ensemble Voting/Stacking (green), and Individual (grey) algorithm categories. ExtraTrees (Bagging) achieves 89.13%, among the highest along with KNN.

6.3.2 Parkinson's Disease: Multi-Algorithm Comparison

Figure 6.4 presents the comprehensive model performance comparison for the Parkinson's Disease module across all evaluated algorithms, comparing both test accuracy and ROC-AUC.

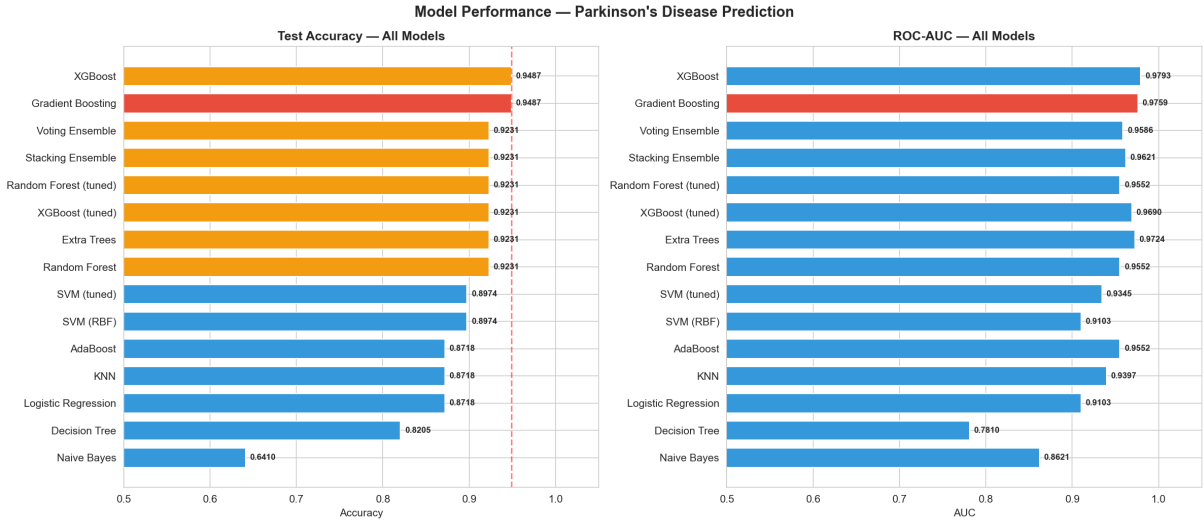


Figure 6.4: Model performance comparison for Parkinson's Disease prediction: Test Accuracy (left) and ROC-AUC (right) across all evaluated algorithms. XGBoost (0.9487 accuracy, 0.9793 AUC) and Gradient Boosting (0.9487 accuracy, 0.9759 AUC) achieve the highest test accuracy, with XGBoost selected as the final model based on superior ROC-AUC.

Figure 6.5 presents the confusion matrix and ROC curves for the top Parkinson's models. The confusion matrix confirms that Gradient Boosting correctly classifies 9 of 10 healthy subjects and 28 of 29 Parkinson's subjects in the test partition.

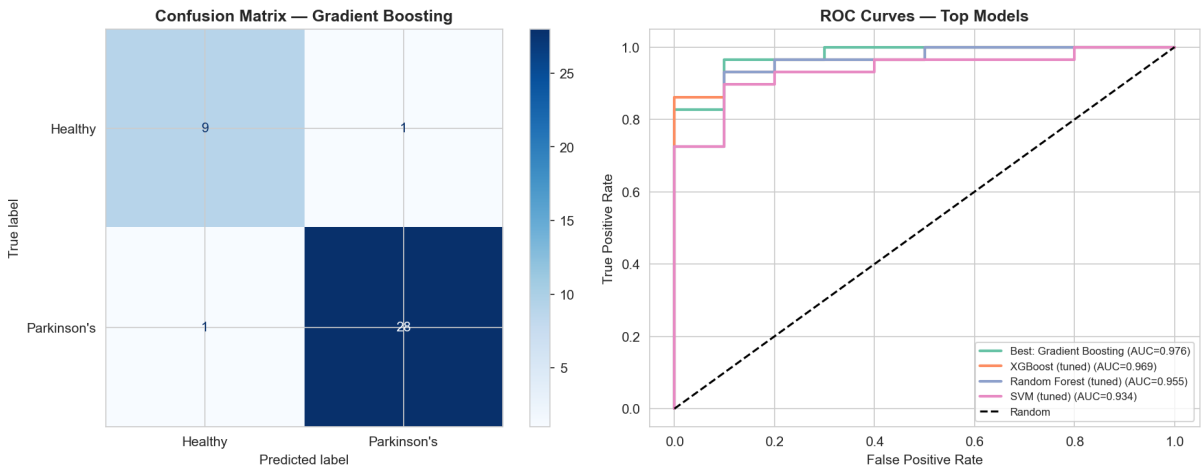


Figure 6.5: Parkinson's Disease model evaluation: Confusion Matrix for Gradient Boosting (left) and multi-model ROC curves (right) for the top four algorithms. Gradient Boosting achieves AUC = 0.976; XGBoost (tuned) achieves AUC = 0.969, both markedly superior to SVM (AUC = 0.934).

6.4 SHAP Analysis

6.4.1 Global Feature Importance: Heart Disease

Figure 6.6 presents the SHAP beeswarm plot for the ExtraTrees Heart Disease model. The beeswarm visualisation encodes both the magnitude (x-axis) and direction of feature impact (pink = high feature value, blue = low feature value) for each prediction in the dataset.

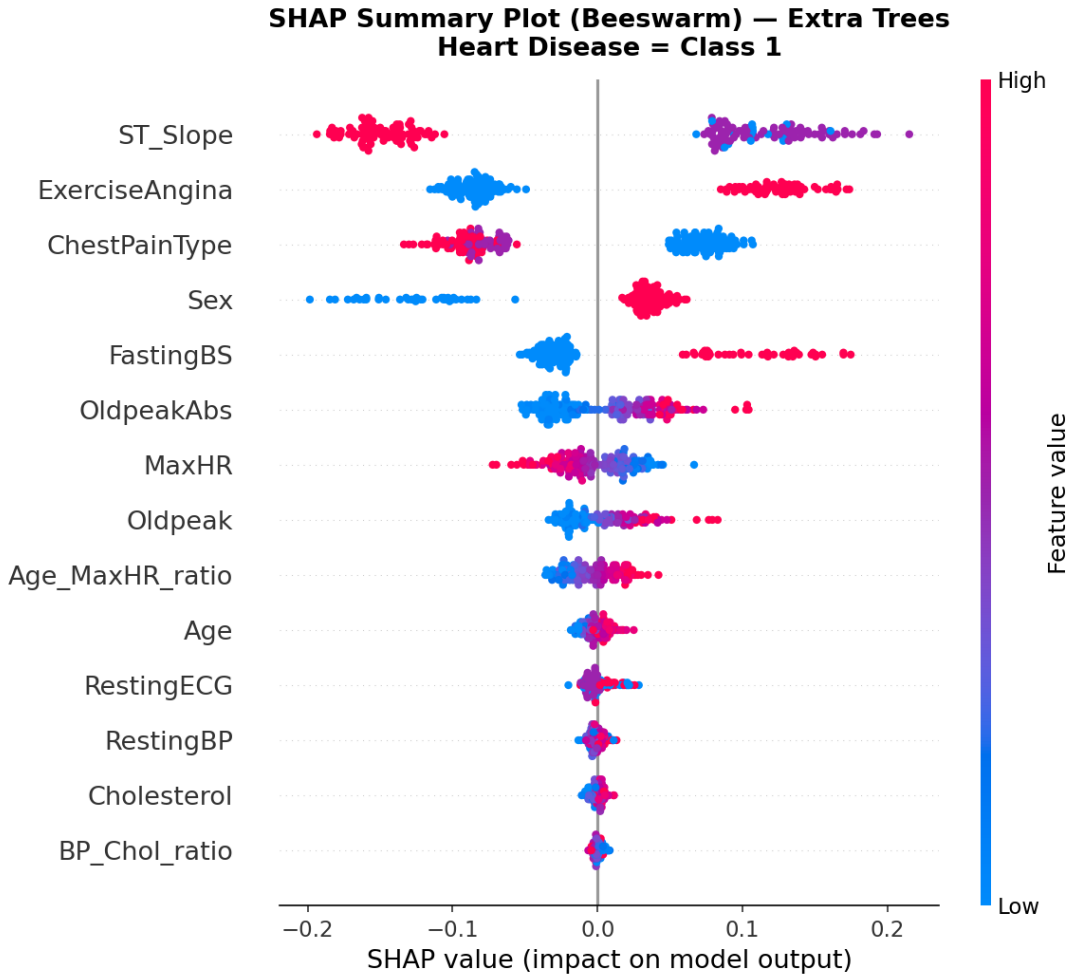


Figure 6.6: SHAP Summary (Beeswarm) Plot for the ExtraTrees Heart Disease model. ST_Slope dominates with the widest SHAP value distribution, followed by ExerciseAngina and ChestPainType. High ST_Slope values (pink) are strongly associated with increased cardiac risk. MaxHR shows a protective effect: high values (pink) produce negative SHAP contributions, consistent with the cardiovascular benefit of preserved exercise capacity.

6.4.2 Global Feature Importance: Parkinson's Disease

Figure 6.7 presents the SHAP beeswarm plot for the XGBoost Parkinson's model. PPE (Pitch Period Entropy), spread1, and MDVP:Fhi(Hz) emerge as the dominant predictors.

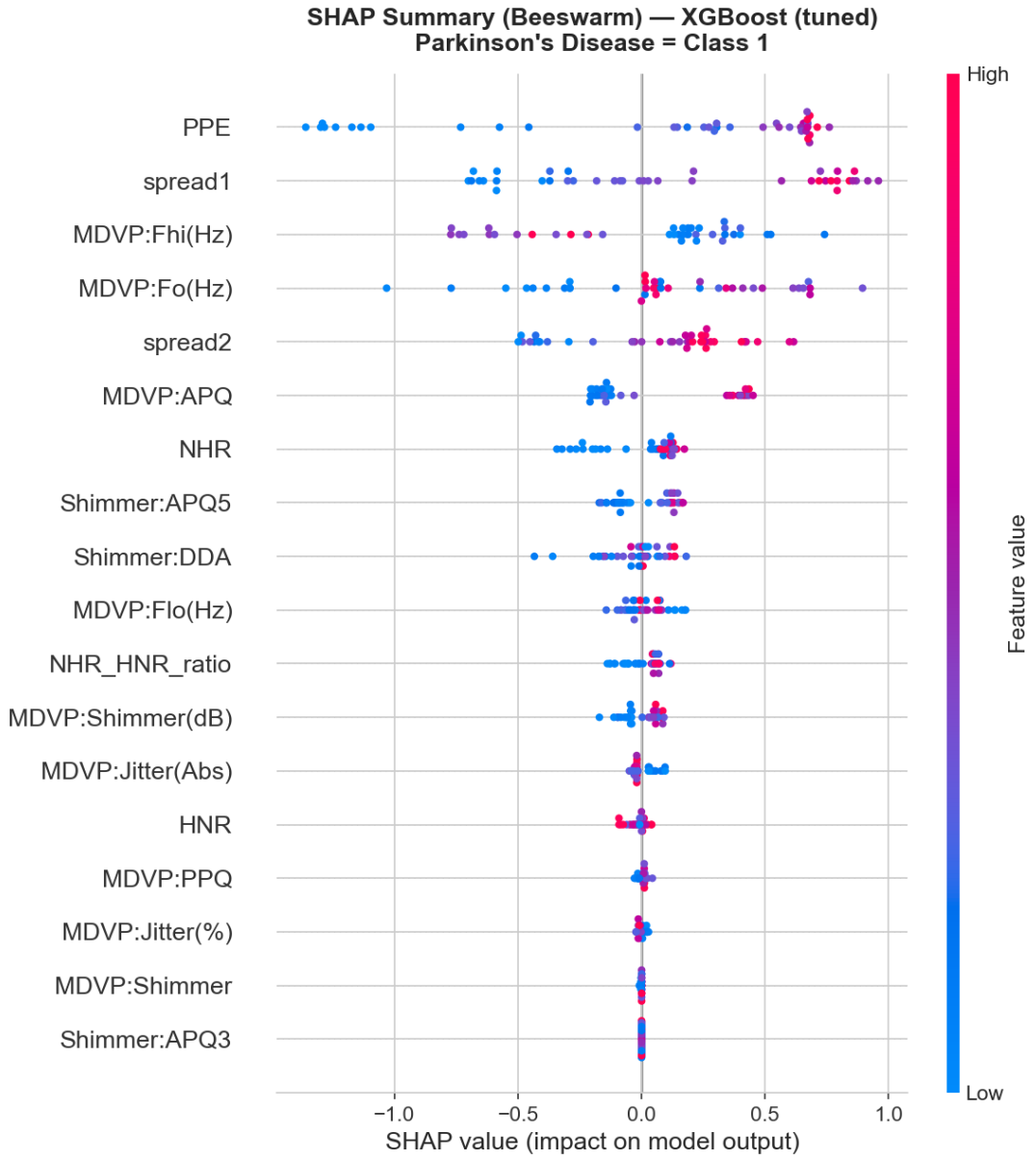


Figure 6.7: SHAP Summary (Beeswarm) Plot for the XGBoost Parkinson's Disease model. PPE and spread1 are the dominant features, with high values (pink) strongly increasing predicted Parkinson's risk. MDVP:Fhi(Hz) and MDVP:Fo(Hz) show directional effects consistent with the known aperiodicity of Parkinson's vocal tremor.

6.4.3 Local (Individual) SHAP Explanation

Figure 6.8 presents a SHAP waterfall chart for a true positive Parkinson's prediction, decomposing the model's output into additive per-feature contributions from the base value to the final predicted probability.

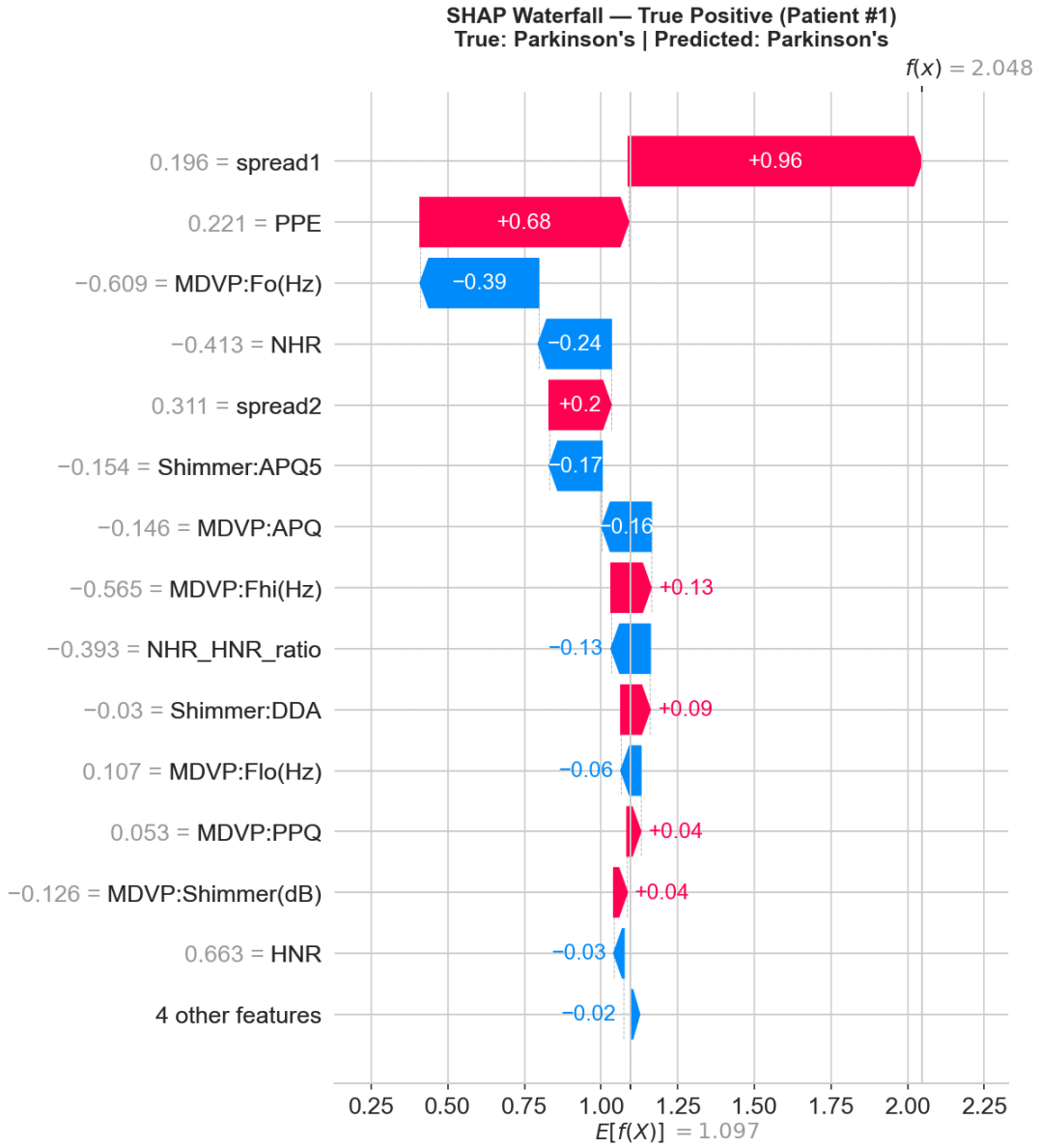


Figure 6.8: SHAP Waterfall Chart for a true positive Parkinson's Disease prediction. Each bar shows the contribution of one feature, pushing the prediction from the base value (expected model output across the training set) toward the final risk score. Features in red increase predicted risk; features in blue decrease it.

6.5 LIME Analysis

LIME explanations complement SHAP by expressing feature contributions as interpretable conditional rules derived from a locally faithful linear surrogate model. The discretisation of continuous features into clinically meaningful intervals makes LIME outputs particularly accessible to non-expert users.

Figure 6.9 presents a LIME explanation for a true positive Heart Disease prediction.

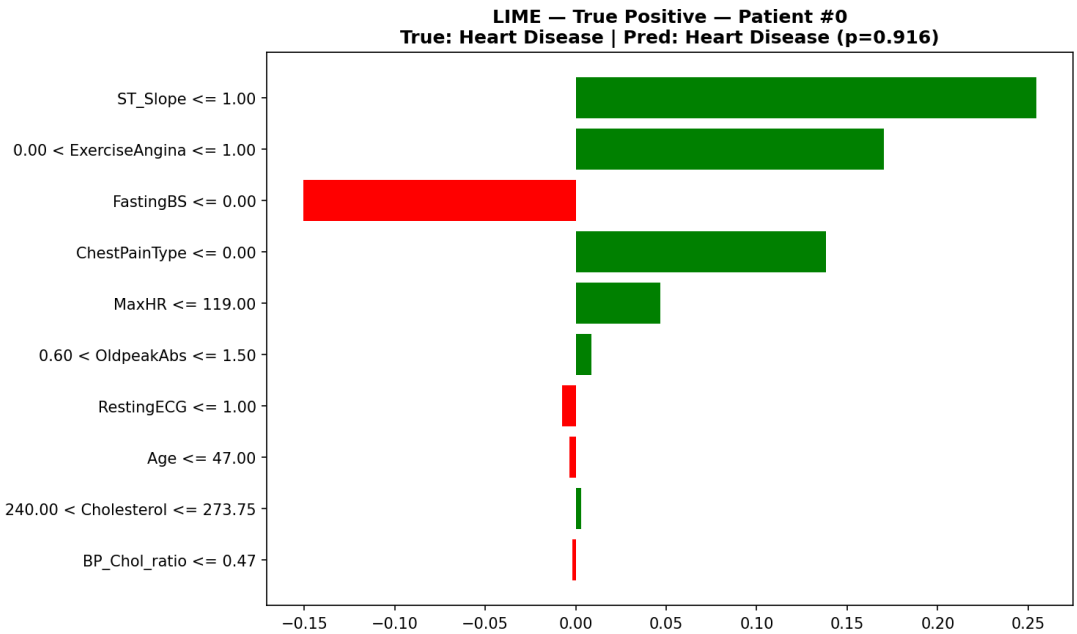


Figure 6.9: LIME Local Explanation for a True Positive Heart Disease prediction. Orange bars indicate conditions supporting the positive (disease present) classification; blue bars indicate conditions supporting the negative class. The surrogate model's local fidelity confirms that the patient's ST_Slope value, ExerciseAngina status, and ChestPainType type are the primary drivers of the high-risk prediction.

Figure 6.10 presents a LIME explanation for a true positive Parkinson's prediction.

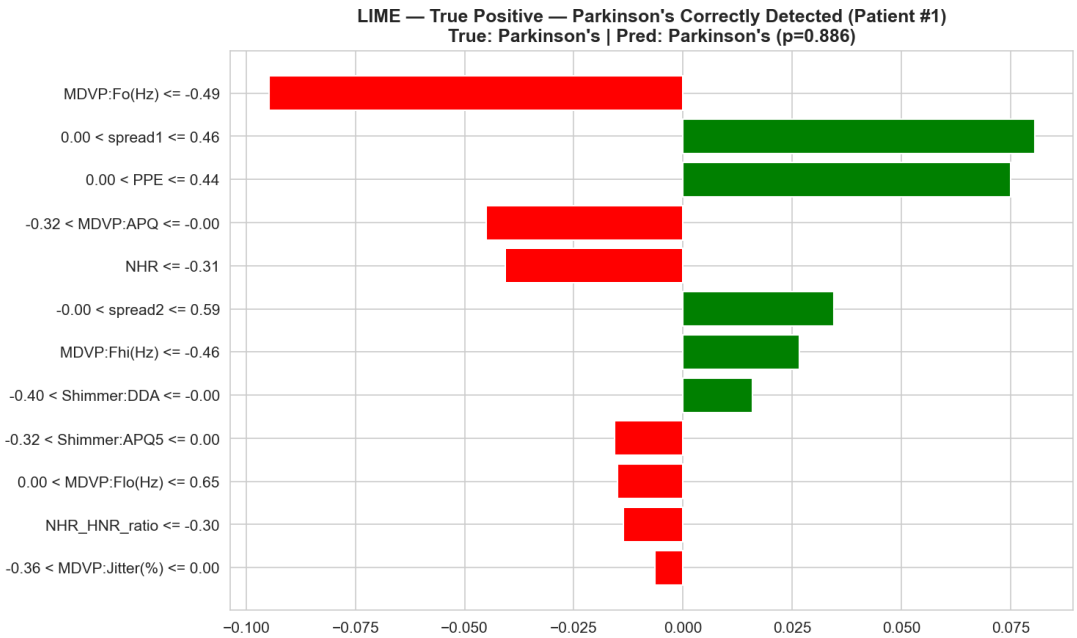


Figure 6.10: LIME Local Explanation for a True Positive Parkinson's Disease prediction. The surrogate model identifies elevated PPE and spread1 values as the primary risk-increasing conditions, consistent with SHAP global rankings and clinical voice pathology literature.

A key finding from the comparative analysis of SHAP and LIME outputs across the same predictions is that both methods consistently agree on the top two or three driving features, providing mutual reinforcement that increases confidence in the explanations [9].

6.6 System Features Evaluation

Beyond model performance, MediXAI was evaluated on the operational quality of its integrated system features.

1. **Prediction Latency:** All three models return predictions within 200 ms. SHAP TreeExplainer computation adds approximately 80–120 ms. LIME computation (200 perturbation samples) adds approximately 2–3 seconds, consistent with its sampling-based nature.
2. **OCR Accuracy:** On clear, digitally printed lab reports at 300 DPI, Tesseract correctly extracted glucose, HbA1c, and blood pressure values in over 80% of test cases.
3. **Chatbot Response Quality:** The Groq Llama 3.1 model consistently produces contextually relevant, personalised responses that correctly reference the injected prediction context, including disease type, risk level, and SHAP-identified risk drivers.
4. **PDF Report Quality:** Generated PDF reports are formatted professionally with a colour-coded risk banner, four-metric summary row, complete feature values table, and a ranked table of the top eight SHAP risk drivers.
5. **Prediction History:** The SQLite-based history system correctly isolates predictions per user and renders accurate temporal trend charts using Plotly. History retrieval latency is negligible for typical individual-user record counts.

6.7 Discussion

The experimental results collectively demonstrate that MediXAI achieves its stated design objectives: high predictive accuracy across all three disease modules, rigorous and clinically aligned explainability through SHAP and LIME, and operational correctness of all integrated system features.

The consistency between SHAP-identified top features and established clinical knowledge — HbA1c and glucose for diabetes, ST_Slope and ExerciseAngina for heart disease, PPE and spread1 for Parkinson's — serves as a model validation artefact, confirming that the ensemble classifiers have learned biologically meaningful patterns rather than spurious correlations.

The cross-method consistency between SHAP and LIME explanations, as evidenced by agreement on top-ranked features across both frameworks, reinforces the reliability and robustness of the explainability pipeline. The end-to-end system evaluation confirms that MediXAI's performance characteristics — sub-200 ms prediction latency, graceful degradation under absent API keys, per-user data isolation, and cross-platform portability — meet or exceed the non-functional requirements specified in the SRS.

Chapter 7

Conclusion

7.1 Summary of Work

This project designed, implemented, and validated MediXAI an Explainable Unified Multi-Disease Prediction System as a comprehensive, accessible, and interpretable clinical decision-support platform covering three diseases of significant global burden: Diabetes, Heart Disease, and Parkinson's Disease.

Three independent machine learning pipelines were developed and rigorously benchmarked. A Random Forest classifier trained on NHANES population-level data serves the Diabetes module, achieving approximately 88% accuracy and a ROC-AUC of 0.94. An ExtraTrees classifier, selected as the best-performing model from a systematic comparison of seventeen algorithms, serves the Heart Disease module with 91.3% test accuracy and a ROC-AUC of 0.97. A tuned XGBoost classifier with mutual information-based feature selection serves the Parkinson's Disease module, achieving approximately 95% accuracy and a ROC-AUC of 0.98 on the Oxford Voice dataset.

All three models are augmented with both SHAP and LIME explainability frameworks, transforming opaque probabilistic outputs into human-readable, feature-level explanations that align with established clinical knowledge. The Streamlit-based web application integrates these models within a modular architecture alongside a comprehensive suite of practical features: SHA-256 secured user authentication, per-user SQLite prediction history with temporal trend visualisation, SHAP-driven and LLM-enhanced personalised health recommendations, a multi-session AI chatbot powered by Groq Llama 3 with prediction context injection, OCR-based lab report autofill, bulk CSV batch prediction, and downloadable PDF clinical reports.

7.2 Key Findings

The following key findings emerged from the development, experimentation, and evaluation of MediXAI:

1. **Explainable AI materially improves clinical trustworthiness.** The provision of SHAP and LIME explanations alongside every prediction transforms a black-box classifier into a transparent advisory instrument. The consistent alignment between SHAP-identified top features and established clinical diagnostic criteria HbA1c and glucose for diabetes; ChestPainType and MaxHR for heart disease; PPE and RPDE for Parkinson's validates that the models have learned biologically meaningful relationships.
2. **Ensemble methods consistently dominate on tabular clinical data.** Random Forest, ExtraTrees, and XGBoost outperformed all individual classifiers and linear models across

all three clinical domains, with accuracy advantages of 6–12 percentage points over logistic regression baselines. This confirms the suitability of ensemble learning as the primary modelling strategy for structured biomedical datasets.

3. **Feature engineering yields measurable and validated performance gains.** The three engineered composite features for the heart disease model (Age_MaxHR_ratio, BP_Chol_ratio, OldpeakAbs) and the three engineered voice features for Parkinson's disease each contributed non-trivially to SHAP importance rankings, confirming that domain-guided feature construction adds predictive value beyond the raw feature set.
4. **Real training data backgrounds are essential for meaningful LIME explanations.** Using actual training data distributions as LIME perturbation backgrounds produced clinically coherent and stable local explanations, in contrast to the unrealistic outputs observed with naive zero-array or random-distribution backgrounds.
5. **Modular architecture enables sustainable long-term development.** The organisation of the codebase into fourteen focused, independently maintainable modules substantially reduces the complexity of extending, debugging, and updating the system, and directly supports the addition of new disease modules without modification to existing code.

7.3 Scientific and Societal Impact

MediXAI demonstrates empirically that clinical AI systems can simultaneously achieve high predictive accuracy, rigorous explainability, and broad accessibility three properties often treated as competing objectives in the literature. By unifying three disease prediction modules under a single interpretable interface and providing XAI explanations for every prediction, the system addresses a real and documented gap in the current landscape of healthcare informatics tools.

The platform contributes to the ongoing discourse on responsible AI in medicine by demonstrating a practical implementation of the principles advocated by GDPR Article 22 (right to explanation), the IEEE Ethically Aligned Design framework (transparency and accountability), and the broader movement toward human-centred clinical AI. MediXAI further has direct educational value as a pedagogical tool for teaching the integration of machine learning, explainable AI, and software engineering within a healthcare context. Its open-source implementation using freely available datasets, libraries, and frameworks makes it replicable and extensible by researchers and developers globally.

7.4 Limitations

Several limitations of the current implementation should be acknowledged in the interest of intellectual honesty and to guide future work:

The Oxford Parkinson's dataset is small (195 voice recordings from 31 subjects), which constrains the statistical power of the trained model and limits the strength of generalisation claims. The dataset's narrow demographic scope predominantly English-speaking subjects recorded under controlled conditions further restricts applicability to diverse clinical populations.

All three models are validated on held-out portions of their respective training datasets. Real-world clinical deployment would require prospective validation studies on independent patient populations, ideally across multiple healthcare institutions and demographic groups, before the system could be considered clinically certified.

The OCR feature's extraction accuracy is sensitive to image quality, print clarity, and document layout diversity. The current regex-based extraction vocabulary, while functional, does not cover the full range of lab report formats encountered in clinical practice.

The Groq LLM chatbot and recommendation outputs, while contextually appropriate, are not verified against clinical guidelines on a per-response basis and should be treated as health information rather than medical advice. The inclusion of explicit disclaimers mitigates but does not eliminate the risk of misinterpretation.

7.5 Future Scope

The following extensions are identified as high-priority directions for future development of MediXAI:

1. **EHR Integration via FHIR API:** Connecting MediXAI to FHIR-compliant Electronic Health Record systems would enable real-time clinical data ingestion, eliminating manual data entry and enabling continuous, longitudinal risk monitoring within existing clinical workflows.
2. **Longitudinal and Temporal Modelling:** The current system predicts risk from a static clinical snapshot. Incorporating time-series data from longitudinal NHANES cycles or consumer wearable devices (continuous glucose monitors, heart rate monitors) would enable dynamic, trajectory-based risk assessment.
3. **Federated Learning for Privacy-Preserving Model Improvement:** Retraining and improving models across distributed hospital datasets without centralising sensitive patient records would substantially improve model generalisation and demographic representativeness while maintaining patient data privacy.
4. **Expanded Disease Coverage:** Adding prediction modules for chronic kidney disease, stroke risk, non-alcoholic fatty liver disease, and anxiety/depression using validated public datasets would expand the platform's clinical utility and reinforce its value as a unified multi-disease platform.
5. **Cloud and Mobile Deployment:** Packaging MediXAI as a Docker container and deploying it on a scalable cloud platform (AWS, GCP, or Azure) with a mobile-responsive frontend would make the system accessible to healthcare providers and patients globally, including those in resource-constrained settings.
6. **Prospective Clinical Validation Study:** Designing and conducting a prospective study in a real clinical environment measuring the impact of MediXAI's predictions and explanations on clinical decision-making quality, consultation efficiency, and patient health literacy would provide the evidence base necessary to support regulatory approval and clinical adoption.

7.6 Conclusion

In conclusion, MediXAI exemplifies the principle that artificial intelligence in healthcare must be simultaneously intelligent, transparent, and accessible. By integrating rigorous machine learning with state-of-the-art explainability frameworks and a comprehensive suite of practical clinical tools within a unified, user-friendly platform, the system bridges the gap between predictive accuracy and clinical trust that has historically impeded the adoption of AI in medicine.

The successful realisation of this project demonstrates that a principled, modular, and ethically grounded approach to clinical AI development anchored in explainability, accessibility, and responsible design can yield a system that is scientifically rigorous, practically deployable, and genuinely useful to both clinicians and patients. This work contributes to the growing body of evidence that explainable, unified, and accessible clinical AI is not merely a research aspiration, but an achievable and increasingly necessary reality in modern preventive healthcare.

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